



ble-sensors-mqtt

Installation and usage manual

Version 1.0.0 - English edition

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> Host support: Linux x86_64/arm64 on Debian/Ubuntu/Raspberry Pi OS and RHEL/Rocky/AlmaLinux/CentOS/Fedora, plus Windows 11+ and macOS Tahoe 26+. Internal and USB Bluetooth are supported through the host OS stack. See `HOSTS.en.md`.

1. Purpose

`ble-sensors-mqtt` 1.0.0 is a multi-sensor gateway for Raspberry Pi. Plugins collect BLE or Tuya Cloud data and emit one common model containing identifier, name, manufacturer, model, protocol, signal strength, timestamp, and all available values. The same snapshot feeds MQTT, Prometheus, and SNMP.

2. Installation

```
sudo apt update
sudo apt install -y bluetooth bluez python3 python3-venv
sudo systemctl enable --now bluetooth
python3 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install '.[all]'
```

On a Pi Zero, use `pip install .` for SwitchBot only, `.[sensors]` for all BLE decoders, or `.[cloud]` for SwitchBot and Tuya. Verify the result with `ble-sensors-mqtt --list-plugins`.

3. Supported plugins

Name	Family	Access
switchbot	SwitchBot	local BLE
xiaomi	Xiaomi/Mijia/HHCC	BLE, optional bind key
govee	Govee	local BLE
inkbird	Inkbird	BLE/connection on compatible models
thermopro	ThermoPro	local BLE
qingping	Qingping/ClearGrass	local BLE
bthome	BTHome, Shelly BLU, ATC/PVVX-BTHome	local BLE
ruuvi	RuuviTag	local BLE
sensornpush	SensorPush	local BLE
airthings	Airthings	active BLE GATT
mopeka	Mopeka	local BLE
tuya-cloud	Tuya/Smart Life	HTTPS cloud

Support varies by model and firmware. Clear-text BLE sensors do not need to be added to the vendor app. The app may coexist with the gateway and can be required for cloud pairing or for obtaining encryption keys.

4. Scan, selection, and aliases

```
.venv/bin/ble-sensors-mqtt --scan --scan-duration 15
.venv/bin/ble-sensors-mqtt --scan --plugin switchbot --plugin bthome
.venv/bin/ble-sensors-mqtt --mqtt-host 192.0.2.10 \
  --device AA:BB:CC:DD:EE:FF \
  --device-name 'AA:BB:CC:DD:EE:FF=Server Room'
```

Scanning needs no MQTT broker. `--device ID` accepts either a BLE MAC or a cloud/plugin identifier. Use `--sensor-name 'ID=NAME'` for Tuya or multi-channel readings. Aliases reach every output while `bluetooth_name` keeps the original radio name.

For encrypted Xiaomi/BTHome frames, store the hexadecimal bind key in a protected file and add `[ble_keys."MAC"], plugin="xiaomi" or "bthome", and bindkey_file="/path/file"` to the

configuration TOML. Never pass keys on the command line.

5. Runtime and MQTT

Home Assistant MQTT Discovery

Add `--home-assistant-discovery` to publish retained Home Assistant discovery configuration. The default discovery prefix is `homeassistant`; override it with `--home-assistant-discovery-prefix`. Each exported sensor becomes one Home Assistant device, with one sensor entity for each scalar payload value plus RSSI and protocol diagnostics. State remains on the regular gateway MQTT state topic and availability is tied to `<mqtt-prefix>/bridge/status`.

```
.venv/bin/ble-sensors-mqtt --mqtt-host 127.0.0.1 --home-assistant-discovery
```

Discovery configuration is retained. Stale discovery topics are explicitly cleared using retained empty payloads so removed sensors/entities disappear from Home Assistant. Keep the runtime state file enabled in long-running/systemd deployments so cleanup also works across restarts.

```
.venv/bin/ble-sensors-mqtt --mqtt-host 127.0.0.1 --once
.venv/bin/ble-sensors-mqtt --mqtt-host 192.0.2.10 --poll-interval 60 --allow-insecure-mqtt
.venv/bin/ble-sensors-mqtt --mqtt-host mqtt.example.net --mqtt-port 8883 \
  --mqtt-username sensor-publisher \
  --mqtt-password-file /etc/ble-sensors-mqtt/mqtt-password --mqtt-tls
```

Polling defaults to 30 seconds. TLS certificates are always verified. Secret files must be 0640 or more restrictive. Every payload contains manufacturer, model, protocol, and all decoder information under data.

6. Tuya Cloud

Run `--cloud-help`, create a Smart Home project in Tuya IoT, and link the Smart Life/Tuya account. Keep Access ID and Access Secret in separate protected files.

```
[cloud.tuya]
enabled = true
region = "eu"
api_key_file = "/etc/ble-sensors-mqtt/tuya-api-key"
api_secret_file = "/etc/ble-sensors-mqtt/tuya-api-secret"
api_device_id = "REFERENCE_DEVICE_ID"
device_ids = ["DEVICE_ID_1"]
```

The TOML file must be 0640 or more restrictive. Test it with:

```
.venv/bin/ble-sensors-mqtt --scan --cloud-config /etc/ble-sensors-mqtt/cloud.toml
```

A Tuya failure does not stop BLE collection. Review cloud privacy, data residency, and service dependency. Tuya devices running BTHome firmware may work locally.

7. Prometheus

Enable the HTTP exporter with:

```
.venv/bin/ble-sensors-mqtt --mqtt-host 192.0.2.10 --prometheus
curl http://127.0.0.1:9105/metrics
```

The default listener is `127.0.0.1:9105`. A non-loopback bind requires the explicit `--allow-external-prometheus` acknowledgement. The endpoint has no built-in authentication or TLS, so external access should be protected by firewall rules, a VPN, or an authenticated reverse proxy.

Exported Prometheus metrics

Metric	Type	Labels	Meaning
ble_sensors_up	gauge	address, name, manufacturer, model, protocol	1 for a fresh snapshot, 0 when the exported snapshot is reused/stale
ble_sensors_stale	gauge	same sensor labels	1 when the snapshot was reused from a previous cycle, otherwise 0
ble_sensors_temperature_celsius	gauge	same sensor labels	recursively discovered numeric temperature value, in degrees Celsius
ble_sensors_humidity_percent	gauge	same sensor labels	recursively discovered numeric humidity value, in percent
ble_sensors_battery_percent	gauge	same sensor labels	recursively discovered numeric battery or battery_percent value
ble_sensors_rssi_dbm	gauge	same sensor labels	top-level Bluetooth RSSI in dBm, when available
ble_sensors_sensor_value	gauge	sensor labels + key, unit	every numeric scalar; booleans are exported as 0/1
ble_sensors_sensor_info	gauge	sensor labels + key, unit, value	every non-null string scalar, represented by a constant sample value of 1
ble_sensors_devices	gauge	none	number of snapshots currently exported, including reused stale snapshots
ble_sensors_cycles_total	counter	none	polling cycles attempted by the gateway
ble_sensors_cycles_failed_total	counter	none	polling cycles marked as failed

The generic metrics are generated only from the normalized sensor data object. Nested dictionaries are flattened with dotted paths such as `air.co2`; list/tuple entries use numeric path components such as `channels.0`. The reserved `data.units` mapping is not exported as sensor data; when available, its leaf-name mapping supplies the unit label. At most 256 scalar values per sensor are exported. `null` values are omitted from Prometheus.

Dedicated temperature/humidity/battery metrics and the generic metric can intentionally expose the same measurement. The dedicated names are convenient for stable dashboards, while the generic family preserves all plugin-provided scalar values. Because `ble_sensors_sensor_info` puts string values in labels, highly variable strings can increase Prometheus cardinality.

Example:

```
ble_sensors_up{address="AA:BB:CC:DD:EE:FF",name="Room",manufacturer="SwitchBot",model="Meter Plus",protocol="SwitchBot"} 1
ble_sensors_temperature_celsius{address="AA:BB:CC:DD:EE:FF",name="Room",manufacturer="SwitchBot",model="Meter Plus",protocol="SwitchBot"} 23.5
ble_sensors_sensor_value{address="AA:BB:CC:DD:EE:FF",name="Room",manufacturer="SwitchBot",model="Meter Plus",protocol="SwitchBot",key="temp",unit="C"} 23.5
```

The same HTTP server also exposes `/healthz` and `/readyz`. `/healthz` reports process liveness; `/readyz` returns HTTP 503 until a polling cycle succeeds and again when the latest successful cycle becomes too old.

```
--prometheus --prometheus-host 0.0.0.0 --prometheus-port 9200 --allow-external-prometheus
```

8. SNMP

Enable the read-only SNMPv2c agent with a protected community file:

```
.venv/bin/ble-sensors-mqtt --mqtt-host 192.0.2.10 --snmp \  
--snmp-community-file /etc/ble-sensors-mqtt/snmp-community
```

The default listener is 127.0.0.1:1161/udp. Non-loopback binding requires `--allow-external-snmp`. SNMPv2c does not encrypt the community or payload, so restrict UDP access or use a VPN/SNMPv3 proxy.

With the default `--snmp-base-oid 1.3.6.1.4.1.32473.1.1`, the exported tree is:

OID suffix	MIB object	Type	Meaning
.1.0	bleSensorsVersion	DisplayString	application identification/version string
.2.0	bleSensorsDeviceCount	Gauge32	number of snapshots currently exported
.10.1.1.I	bleSensorsIndex	Integer32	transient device row index
.10.1.2.I	bleSensorsAddress	DisplayString	normalized sensor identifier/address
.10.1.3.I	bleSensorsName	DisplayString	effective sensor name/alias
.10.1.4.I	bleSensorsRssi	Integer32	RSSI in dBm, omitted when unavailable
.10.1.5.I	bleSensorsTemperatureMilliCelsius	Integer32	temperature multiplied by 1000
.10.1.6.I	bleSensorsHumidityMilliPercent	Gauge32	relative humidity multiplied by 1000
.10.1.7.I	bleSensorsBatteryPercent	Gauge32	battery percentage rounded to an integer
.10.1.8.I	bleSensorsObservedAt	DisplayString	original observation timestamp
.10.1.9.I	bleSensorsManufacturer	DisplayString	manufacturer, or Unknown
.10.1.10.I	bleSensorsModel	DisplayString	model, or Unknown
.10.1.11.I	bleSensorsProtocol	DisplayString	protocol, or Unknown
.10.1.12.I	bleSensorsStale	Gauge32	1 for reused/stale data, otherwise 0
.20.1.1.I.J	bleSensorsValueKey	DisplayString	flattened scalar key/path
.20.1.2.I.J	bleSensorsValue	DisplayString	scalar rendered as text
.20.1.3.I.J	bleSensorsValueType	DisplayString	null, boolean, number, or string
.20.1.4.I.J	bleSensorsValueUnit	DisplayString	unit from data.units, when available
.20.1.5.I.J	bleSensorsValueDeviceIndex	Integer32	device index I
.20.1.6.I.J	bleSensorsValueIndex	Integer32	scalar row index J

Rows in .10 are ordered by the normalized sensor identifier, so index I is transient and may change when the exported device set changes. Optional common-measurement columns are absent when their values are not available. Table .20 contains up to 256 scalar values from each sensor data object and uses the same flattening rules as Prometheus. SNMP textual fields are capped by the agent at 512 encoded bytes.

The bundled textual MIB is docs/BLE-SENSORS-MQTT-MIB.txt. PEN 32473 is documentation-only. In production, use an assigned enterprise OID with --snmp-base-oid. Changing the base OID relocates the same suffix layout shown above; the bundled textual MIB itself still names the documented default root.

Example walks:

```
snmpwalk -v2c -c RANDOM_COMMUNITY 127.0.0.1:1161 1.3.6.1.4.1.32473.1.1
snmpwalk -v2c -c RANDOM_COMMUNITY 127.0.0.1:1161 1.3.6.1.4.1.32473.1.1.10
snmpwalk -v2c -c RANDOM_COMMUNITY 127.0.0.1:1161 1.3.6.1.4.1.32473.1.1.20
```

9. systemd

For a complete installation starting from a GitHub clone (host prerequisites, Bluetooth check, venv, and systemd):

```
git clone https://github.com/desalvo/ble-sensors-mqtt.git
cd ble-sensors-mqtt
sudo scripts/install-from-github.sh --mqtt-host mqtt.example.net --prometheus
```

Use the systemd installer directly when the clone and prerequisites already exist:

```
# Interactive; CLI values are the defaults shown by the prompts
sudo scripts/install-systemd.sh --mqtt-host mqtt.example.net --prometheus

# Fully unattended provisioning
sudo scripts/install-systemd.sh --non-interactive \
  --mqtt-host mqtt.example.net --mqtt-tls \
  --mqtt-username ble-sensors-publisher --mqtt-password-file ./mqtt-password \
  --home-assistant-discovery --prometheus
```

The installer creates the service user, isolated venv, root-owned configuration and /etc/ble-sensors-mqtt/config.toml; repeatable options are stored as TOML arrays and values containing spaces are preserved without shell parsing. In interactive mode, CLI values remain the defaults shown by the wizard and already supplied repeatable options are retained. Use --non-interactive to install only from CLI/default values. See docs/SYSTEMD.en.md.

Windows/macOS graphical installation and settings

Tagged Windows releases include a standard graphical -setup.exe installer; tagged macOS releases include standard .pkg installers for Apple Silicon and Intel. Both install a graphical **ble-sensors-mqtt Settings** application. Windows uses a system Windows Service and %ProgramData%\ble-sensors-mqtt\config.toml; macOS uses a per-user launchd LaunchAgent and ~/Library/Application Support/ble-sensors-mqtt/config.toml. The GUI can save configuration, start/stop/restart the service, test Bluetooth, and open the configuration folder. CLI arguments override TOML without rewriting it. See docs/NATIVE-INSTALLERS.en.md and docs/CONFIGURATION.en.md.

10. Docker and Kubernetes

```
# Multiarch build + Docker Hub push of desalvo/ble-sensors-mqtt:<VERSION>
docker login
scripts/build-docker.sh

# Continuous Docker service
cd docker
cp .env.example .env
cp arguments.example arguments
docker compose up -d

# Continuous Kubernetes service
kubectl label node NODE_NAME ble-sensors-mqtt/bluetooth=true
kubectl apply -k kubernetes/
```

Before Docker/Kubernetes, validate the host with sudo scripts/check-bluetooth-host.sh --strict. Both mount /run/dbus read-only and use host BlueZ over system D-Bus; privileged mode is not required in the supported model. Kubernetes also includes kubernetes/bluetooth-test-pod.yaml to validate --scan on the selected node.

The image supports linux/amd64 and linux/arm64. Docker/Kubernetes mount the host system D-Bus for BlueZ, persist runtime state, send MQTT outbound, expose Prometheus on TCP/9105 and SNMP on UDP/1161. Optional Kubernetes LoadBalancer manifests expose Prometheus and SNMP externally when explicitly applied. scripts/build-docker.sh pushes desalvo/ble-sensors-mqtt:<VERSION> by default; CI can publish the

versioned image on vX.Y.Z tags and latest on main. See docs/DOCKER.en.md and docs/KUBERNETES.en.md; external monitoring endpoints must be restricted to trusted networks.

11. Extensions

A third-party local package implements `SensorPlugin.decode()`, returns `SensorReading` objects, and registers itself in the `ble_sensors_mqtt.sensor_plugins` entry-point group. A third-party cloud provider implements `CloudPlugin.poll()` and registers a factory/class in `ble_sensors_mqtt.cloud_plugins`; its constructor receives (`config`, `read_secret`) and its default TOML section is `[cloud.<entry-point-name-without--cloud>]`. Every reading must contain manufacturer, model, and protocol. Plugin exceptions are isolated.

12. Security and troubleshooting

- do not run as root;
- protect MQTT, Tuya, bind-key, and SNMP files;
- use minimal MQTT ACLs and TLS;
- do not expose Prometheus/SNMP without network controls;
- update and audit dependencies with `pip-audit`;
- test actual sensor models before production.

```
bluetoothctl show
.venv/bin/ble-sensors-mqtt --list-plugins
.venv/bin/ble-sensors-mqtt --scan --scan-duration 20 --log-level DEBUG
journalctl -u ble-sensors-mqtt --since today
```

The AgID review is a technical self-assessment, not a certification.

13. Build and rollback

`scripts/build-package.sh` generates a UTC build ID in YYYYMMDD-HHMM format, ZIP, TAR.GZ, and SHA-256 checksums. Keep the previous package for rollback; do not modify site-packages in place.

14. Production health and release gates

Health and readiness

When Prometheus is enabled, the same HTTP server provides:

- `/metrics` for Prometheus;
- `/healthz` for process liveness;
- `/readyz` for readiness. It returns HTTP 503 until one polling cycle succeeds and again if successful polling becomes stale.

Non-loopback Prometheus/health exposure requires `--allow-external-prometheus`; non-loopback SNMPv2c requires `--allow-external-snmp`. Remote or credentialed plaintext MQTT requires `--allow-insecure-mqtt`. These overrides are deliberate risk acknowledgements, not recommended production defaults.

`--plugin-timeout` limits one decoder/cloud call (default 15 seconds). `--stale-cycles` controls when retained MQTT state for a missing sensor is cleared (default 3 consecutive cycles). `--state-file` persists this cleanup state across restarts; the supplied systemd unit uses `/var/lib/ble-sensors-mqtt/state.json`.

Release verification

Install `.[all,dev,release]` and run:

```
scripts/release-check.sh
```

The gate requires lint, tests/coverage, Bandit, dependency audit, EN/IT PDF generation, wheel and sdist build, `twine check`, CycloneDX SBOM generation, and release package verification. CI repeats runtime tests on Python

3.11, 3.12 and 3.13.

For GitHub publication, push the source commit first and then push a matching `vX.Y.Z` tag. The tag workflow runs all gates, builds deterministic release assets, creates consolidated SHA-256 checksums, and publishes the GitHub Release automatically only after the test and build jobs succeed. The release job checks that the tag version matches `VERSION` and `pyproject.toml`.

Stale fallback and persistent MQTT cache

Use `--reuse-stale-data` to preserve the last in-memory sensor snapshot when a cycle misses that sensor or produces no sensor data. Reused snapshots retain the original observation timestamp and set `stale: true`; fresh readings set `stale: false`. The MQTT SQLite spool is enabled by default, has a 1 GiB logical limit, continues collecting while the broker is offline, flushes FIFO after reconnection, and removes records only after MQTT acknowledgement. Configure it with `--mqtt-cache-path PATH`, `--mqtt-cache-max-size SIZE`, or `--no-mqtt-cache`.

Optional web frontend

Enable the authenticated web console with `--frontend` after installing `. [web]` or `. [all]`. Default bind is `127.0.0.1:8080`; use `--allow-external-frontend` for a non-loopback address and protect remote access with HTTPS or a TLS reverse proxy. Initial credentials are `admin / password`, with mandatory password change at the first login. The console provides responsive desktop/mobile sensor status, admin/reader authorization, persistent runtime settings, local/LDAP/OIDC authentication, TOTP MFA for local/LDAP users, user administration and encrypted import/export of configuration plus available cache/state. Full details are in `docs/FRONTEND.en.md`.